

SANKEETH GANESWARAN

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EDUCATION

University of Waterloo
Bachelor of Computer Science

2026

EXPERIENCE

Engine Programmer (Software Engineer Intern)

May 2025 – Aug. 2025

Epic Games

Cary, NC (Remote)

- Architected and optimized **distributed network replication systems** in **Unreal Engine 5**, handling **real-time state synchronization** across thousands of concurrent game clients using custom **Iris replication protocol**.
- Debugged and resolved critical **engine-level performance bottlenecks** in **C++**, including **RPC serialization** and **stable name subobject replication**, improving test reliability by **40%** and reducing network overhead.
- Designed and implemented **internal automation tooling** including replay analysis systems to ensure **backward compatibility** across game engine versions, reducing regression testing time by **50%**.
- Built a dynamic **bandwidth allocation system** to prioritize and budget actor replication traffic; reduced over-allocation stalls by **30%** in high-load scenarios and improved fairness across **10k+ simultaneous connections**.
- Wrote unit and integration tests and implemented monitoring to ensure safe production rollouts.

Software Engineer Intern

Jan. 2025 – April 2025

X (formerly Twitter)

Palo Alto, CA

- Led end-to-end migration of **authentication infrastructure** from SMS-based 2FA to **passkey authentication**, impacting **250k+ daily active users** and improving security posture across the platform.
- Designed and implemented **full-stack authentication flows** using **Scala microservices** (backend), **React** (web), and native **iOS/Android** clients, collaborating with product and security teams to define rollout and testing strategy.
- Optimized cross-domain authentication flow between **twitter.com** and **x.com**, eliminating redundant redirects and reducing authentication latency by **50%**, improving user experience for millions of users.
- Collaborated with infrastructure and security teams to ensure **FIDO2 compliance** and designed comprehensive rollback mechanisms for production deployment safety.

Game Developer (Software Engineer Intern)

Sept. 2023 – Aug. 2024

ArenaX Labs

Toronto, ON

- Built core gameplay systems and **Python-based ML training infrastructure** for **AI Arena**, a competitive platform fighter serving **200k+ monthly active users** with **real-time multiplayer** capabilities.
- Implemented physics-based combat systems in **JavaScript/ThreeJS**, including **finite state machines** and **20+ real-time visual effects**, while optimizing engine performance to reduce memory footprint by **40%** and improve load times by **35%**.

Gameplay Programmer (Software Engineer Intern)

Jan. 2023 – April 2023

Lucky VR

Toronto, ON

- Developed **cross-platform VR gameplay features** in **Unity/C#** for **PokerStars VR** across PC, Quest, and PSVR, implementing scalable prop and avatar systems supporting **50+ customizable items** with optimized **3D rendering** and physics interactions.
- Fixed **70+ production bugs** and refactored systems to reduce latency by **60%** using **Photon PUN RPC** for synchronization.

Autonomous Vehicle Android Developer (Software Engineer Intern)

May 2022 – Aug. 2022

Ford Motor Company of Canada

Oakville, ON

- Built **Android automotive application** in **Kotlin** to interface with vehicle hardware via **CAN bus** and **SOA messaging**, controlling **60+ vehicle property functions** including climate, lighting, and diagnostics.

PROJECTS

Creature Clash - Distributed Multiplayer Game | Unity, C#, Photon PUN

- Architected physics-based mobile game with **client-server architecture**, implementing **lag compensation** and **client-side prediction** for smooth gameplay under varying network conditions.

MeetKicker Chrome Extension | JavaScript, Chrome APIs

- Published browser extension to **Chrome Web Store** with **250+ active users**, automating Google Meet session management using DOM manipulation and event-driven programming.

TECHNICAL SKILLS

Languages: C++, C#, JavaScript, TypeScript, Python, Java, Kotlin, C, SQL, Scala, Go, Swift, R

Technologies: AWS, Docker, Kubernetes, Git, Linux, Redis, PostgreSQL, TensorFlow, React, Node.js, Unity, Unreal Engine

Concepts: Distributed Systems, Microservices, Network Programming, Real-time Systems, Machine Learning